Rakesh Singh

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SUMMARY

AI Engineer with **4 years** of production backend experience, building **LLM, RAG and AI agent** systems in **Python** (LangGraph, MCP), backed by **Java / Spring Boot** microservices that handled **2 million** daily transactions. Ships AI features that answer accurately from company data, stay secure and keep costs low on **OpenAI, Anthropic Claude and AWS Bedrock**. Creator of **toolism**, an open-source library that cuts LLM token costs.

OPEN SOURCE

toolism — open-source LLM cost optimizer | Python, MCP, PyPI github.com/rakeshnitb23/toolism

- Built toolism, an open-source **Python** library and CLI that shrinks the tool / **MCP** definitions AI agents resend on every turn, cutting those tokens by **38%** across **12** public MCP servers and lowering LLM costs.
- Made every optimization safe to adopt: automated property-based tests (**pytest**, Hypothesis) prove each rewritten tool definition behaves exactly like the original, so teams cut costs without breaking their agents.
- Shipped a **GitHub Action** that blocks pull requests that would push AI token costs over budget, giving teams a **CI/CD** cost guardrail; published on **PyPI** with **54** stars and **28** contributors.

EXPERIENCE

AI Backend Engineer, Genpact Feb 2026 – Sep 2026

GroundedDocs — AI assistant that answers from a company's private documents with the exact source, or says "I don't know" instead of guessing.

- Built a secure **RAG** search engine on **PostgreSQL** and **pgvector** (hybrid semantic + keyword search with reranking) that finds the right passage in the top 5 results **92%** of the time, while keeping each customer's data private.
- Delivered cited, structured **LLM** answers where **94%** of citations matched the source, and the assistant correctly refused **86%** of questions its documents couldn't answer, reducing hallucination risk for business users.
- Built a **Python** document-processing pipeline that re-processes only changed files, so users always get answers from the latest version of their documents, with no duplicate or outdated sources.
- Secured the AI assistant against **prompt injection**, data leaks between customers and malicious PDFs through threat modeling and a **30+** case attack test suite every release had to pass (**97%** pass rate).
- Built an OpenAI-compatible **LLM gateway** over **OpenAI, Anthropic Claude and AWS Bedrock** with streaming, per-customer API keys, token budgets, retries and caching, keeping AI features stable under simulated failures at **$0.004** per successful request.
- Built automated AI quality checks (**evals** with Ragas, DeepEval and **Langfuse** monitoring) that block a release if accuracy, security, speed or cost gets worse, so every update is safe to ship.
- Added a **LangGraph** AI agent with **MCP** tool access for complex questions standard search couldn't answer, with strict step and permission limits, kept only where it improved answer quality without raising cost or latency.

Backend Java Developer, Genpact (Client: Shutterfly, USA) Oct 2022 – Jan 2026

- Built and scaled **Java 11 / Spring Boot REST microservices** handling **2 million** daily transactions for Shutterfly's photo platform, cutting API response times by **65%**.
- Kept **15+** microservices online **99.9%** of the time with **Resilience4j** fault tolerance, even when dependencies failed.
- Tuned **PostgreSQL, MongoDB and Redis** so data loaded **70%** faster and costs dropped **30%**.
- Built a photo-upload pipeline for **5 million** uploads a day in under **200 ms**; smart compression cut storage costs by **40%**.
- Built **React.js + TypeScript** internal tools and paginated **REST APIs** over **100k+** records, making search **60%** faster.
- Automated release management with a **TypeScript** service, cutting manual coordination by **85%**.
- Mentored **3** developers in test-driven development (**JUnit 5, Mockito**), raising code coverage from **65%** to **92%**.

TECHNICAL SKILLS

GenAI & Agents	**LLMs**, **RAG**, **AI agents**, **LangGraph**, LangChain, **MCP**, tool calling, subagents, structured outputs
LLM Platforms	**OpenAI**, **Anthropic Claude**, **AWS Bedrock** (Knowledge Bases, Guardrails), AgentCore
RAG & Search	**Hybrid search** (pgvector + BM25 + RRF), cross-encoder reranking, citation grounding, agent memory
Evals & Observability	**Eval gates**, golden datasets, faithfulness, LLM-as-judge, **Langfuse**, Ragas, DeepEval, OpenTelemetry
LLMOps & Cost	**LLM gateway**, token budgets, caching, retries, circuit breakers, cost per successful task, model routing
AI Security	Threat modeling (STRIDE), **prompt-injection defense**, adversarial testing, fail-closed auth, least privilege
Backend	**Python (FastAPI)**, **Java 8/11/17 (Spring Boot)**, **REST APIs**, microservices, TypeScript, JavaScript, React.js
Data	**PostgreSQL**, **pgvector**, **SQL**, MongoDB, Redis, DynamoDB, query optimization
Cloud & DevOps	**AWS** (EC2, RDS, S3, Bedrock, Lambda), Docker, **Kubernetes**, Terraform, CI/CD (Jenkins, GitHub Actions)
Testing & Tools	pytest, Hypothesis, JUnit 5, Mockito, TDD, Git, Claude Code, Cursor

EDUCATION

National Institute of Technology (NIT) Bhopal Jun 2018 – Jun 2022

Bachelor of Technology (B.Tech)

ADDITIONAL

- **Leadership:** Led ISTE chapter initiatives at NIT Bhopal, organizing technical events for 200+ students; won 2 internal hackathons.
- **Community:** Volunteered 100+ hours at Arushi NGO, Bhopal, working with specially abled children.